

Docker Fundamentals: Working with Images and Containers

To Begin with the Lab

Summary of the Lab

This lab demonstrates how to deploy an NGINX web server using Docker on an Ubuntu EC2 instance. It covers installing and verifying Docker, starting the Docker service, pulling and running the NGINX container, publishing port 80, accessing the web server through the EC2 public IP, managing containers in foreground and background modes, inspecting Docker images and layers, and finally stopping, removing containers, and deleting the NGINX image.

Prerequisites

- AWS Account
- Ubuntu EC2 Instance
- Security Group allowing:
 - SSH (22)
 - HTTP (80)
- SSH access to EC2
- Now launch an **Ubuntu EC2 instance**.
- **Verify Docker is Installed**
- Check the Docker version:

```
docker --version
```

- If Docker is installed, you'll see the version information.

```
ubuntu@ip-172-31-45-84:~$ docker --version
Command 'docker' not found, but can be installed with:
sudo apt install docker.io      # version 29.1.3-0ubuntu4.1, or
sudo apt install podman-docker  # version 5.7.0+ds2-3build1
ubuntu@ip-172-31-45-84:~$
```

- So, docker is not installed, let's install the docker first
- Run the commands given below:

```
sudo apt update
sudo apt install -y docker.io
```

- Now run the command

```
docker --version
```

```
ubuntu@ip-172-31-45-84:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
ubuntu@ip-172-31-45-84:~$
```

- Verify the Docker Service

- Check whether Docker is running:

```
sudo systemctl status docker
```

- If it is not running:

```
sudo systemctl start docker
```

- Enable Docker to start automatically after reboot:

```
sudo systemctl enable docker
```

```
ubuntu@ip-172-31-45-84:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-07-07 09:11:33 UTC; 4min 1s ago
 Invocation: 3837a6f112ec48f5af1aae93ee5f5332
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 2318 (dockerd)
      Tasks: 0
     Memory: 59M (peak: 59.8M)
        CPU: 400ms
    CGroup: /system.slice/docker.service
            └─2318 /usr/bin/dockerd -B fd:// --containerd=/run/containerd/containerd.sock

Jul 07 09:11:33 ip-172-31-45-84 dockerd[2318]: time="2026-07-07T09:11:33.466759754Z" level=info msg="Restoring containers: start."
Jul 07 09:11:33 ip-172-31-45-84 dockerd[2318]: time="2026-07-07T09:11:33.522108189Z" level=info msg="Deleting nftables IPv4 rules" error="exit status 1"
Jul 07 09:11:33 ip-172-31-45-84 dockerd[2318]: time="2026-07-07T09:11:33.527356957Z" level=info msg="Deleting nftables IPv6 rules" error="exit status 1"
Jul 07 09:11:33 ip-172-31-45-84 dockerd[2318]: time="2026-07-07T09:11:33.855448743Z" level=info msg="Loading containers: done."
Jul 07 09:11:33 ip-172-31-45-84 dockerd[2318]: time="2026-07-07T09:11:33.878885303Z" level=info msg="Docker daemon" commit=29.1.3-0ubuntu4.1 containerd-snapshotter=
Jul 07 09:11:33 ip-172-31-45-84 dockerd[2318]: time="2026-07-07T09:11:33.879695992Z" level=info msg="Initializing buildkit"
Jul 07 09:11:33 ip-172-31-45-84 dockerd[2318]: time="2026-07-07T09:11:33.937976066Z" level=info msg="Completed buildkit initialization"
Jul 07 09:11:33 ip-172-31-45-84 dockerd[2318]: time="2026-07-07T09:11:33.954514528Z" level=info msg="Daemon has completed initialization"
Jul 07 09:11:33 ip-172-31-45-84 systemd[1]: Started docker.service - Docker Application Container Engine.
Jul 07 09:11:33 ip-172-31-45-84 dockerd[2318]: time="2026-07-07T09:11:33.954716550Z" level=info msg="API listen on /run/docker.sock"
lines 1-23/23 (END)
```

- Run the NGINX Container
- Run the following command:

```
sudo docker run -d -p 80:80 nginx
```

- Command explanation:
 - docker run → Creates and starts a container.
 - -d → Runs the container in the background.
 - -p 80:80 → Maps port 80 on the EC2 instance to port 80 inside the container.
 - nginx → Downloads (if necessary) and runs the official NGINX image.

```
ubuntu@ip-172-31-45-84:~$ sudo docker run -d -p 80:80 nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
df68ee7e7a00: Pull complete
acf093e7a04f: Pull complete
1b30016634d5: Pull complete
e95a6c7ea7d4: Pull complete
1645c1e06f46: Pull complete
cd9307c9ecd8: Pull complete
fcb6fd84b2a0: Pull complete
1cf7d051b485: Download complete
e2c07e54e55a: Download complete
Digest: sha256:ec4ed8b5299e5e90694af7750eb6dffd2627317d30544d056b0371f8082f7bce
Status: Downloaded newer image for nginx:latest
b2a608dcde38cdc8bb558de30bb60e299c40c4a9450e17014ed2d52ffcf5582
ubuntu@ip-172-31-45-84:~$
```

- Docker Downloads the Image
- If the image is not already available:
- Docker connects to Docker Hub.
- Downloads the NGINX image.
- Stores it locally.
- Creates and starts the container.

- Verify the Container Is Running
- List running containers:

```
sudo docker ps
```

- Expected output (container ID will differ):

```
ubuntu@ip-172-31-45-84:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS                               NAMES
b2a608dcede3   nginx    "/docker-entrypoint..." 3 minutes ago    Up 3 minutes    0.0.0.0:80->80/tcp, [::]:80->80/tcp    keen_leakey
ubuntu@ip-172-31-45-84:~$
```

- Access the NGINX Web Server
- Open a browser and visit:
- <http://<EC2-Public-IP>>
- You should see the default **Welcome to nginx!** page.

Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

- Stop the Container
- List containers:

```
sudo docker ps
```

- Stop the container:

```
sudo docker stop <container-id>
```

```
ubuntu@ip-172-31-45-84:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS                               NAMES
b2a608dcede3   nginx    "/docker-entrypoint..." 8 minutes ago    Up 8 minutes    0.0.0.0:80->80/tcp, [::]:80->80/tcp    keen_leakey
ubuntu@ip-172-31-45-84:~$ sudo docker stop b2a608dcede3
b2a608dcede3
ubuntu@ip-172-31-45-84:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS    NAMES
ubuntu@ip-172-31-45-84:~$
```

Running a Docker Container in Foreground vs Background

- Verify the NGINX Image
- List the images available on your EC2 instance:

```
sudo docker images
```

- Run the Container in the Foreground
- Run NGINX Without -d
- `sudo docker run nginx`

```

ubuntu@ip-172-31-45-84:~$ sudo docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
nginx:latest	ec4ed8b5299e	241MB	66MB	📄

```

ubuntu@ip-172-31-45-84:~$ sudo docker run nginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/07/07 09:31:42 [notice] 1#1: using the "epoll" event method
2026/07/07 09:31:42 [notice] 1#1: nginx/1.31.2
2026/07/07 09:31:42 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/07/07 09:31:42 [notice] 1#1: OS: Linux 7.0.0-1006-aws
2026/07/07 09:31:42 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:524288
2026/07/07 09:31:42 [notice] 1#1: start worker processes
2026/07/07 09:31:42 [notice] 1#1: start worker process 29
2026/07/07 09:31:42 [notice] 1#1: start worker process 30

```

Publishing Container Ports with -p

- Run the Container in Detached Mode
- Start NGINX in the Background
- Run:

```
sudo docker run -d nginx
```

```

ubuntu@ip-172-31-45-84:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS    NAMES
ubuntu@ip-172-31-45-84:~$ sudo docker run -d nginx
2198d760ba3f554c96aef261b85f48f8f9996fb25997fbd7671775ae23254723
ubuntu@ip-172-31-45-84:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS    NAMES
2198d760ba3f   nginx     "/docker-entrypoint..." 6 seconds ago    Up 5 seconds    80/tcp    nifty_davinci
ubuntu@ip-172-31-45-84:~$

```

Understanding Docker Images

- View All Containers
- To display both running and stopped containers:

```
sudo docker ps -a
```

```

ubuntu@ip-172-31-45-84:~$ sudo docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS    NAMES
2198d760ba3f   nginx     "/docker-entrypoint..." 4 minutes ago    Up 4 minutes    80/tcp    nifty_davinci
1b30746055aa   nginx     "/docker-entrypoint..." 8 minutes ago    Exited (0) 7 minutes ago    lucid_darwin
b2a608dc3     nginx     "/docker-entrypoint..." 23 minutes ago    Exited (0) 14 minutes ago    keen_leakey
ubuntu@ip-172-31-45-84:~$

```

```

ubuntu@ip-172-31-45-84:~$ sudo docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS    NAMES
2198d760ba3f   nginx     "/docker-entrypoint..." 4 minutes ago    Up 4 minutes    80/tcp    nifty_davinci
1b30746055aa   nginx     "/docker-entrypoint..." 8 minutes ago    Exited (0) 7 minutes ago    lucid_darwin
b2a608dc3     nginx     "/docker-entrypoint..." 23 minutes ago    Exited (0) 14 minutes ago    keen_leakey
ubuntu@ip-172-31-45-84:~$ sudo docker stop 2198d760ba3f
2198d760ba3f
ubuntu@ip-172-31-45-84:~$ sudo docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS    NAMES
2198d760ba3f   nginx     "/docker-entrypoint..." 7 minutes ago    Exited (0) 6 seconds ago    nifty_davinci
1b30746055aa   nginx     "/docker-entrypoint..." 10 minutes ago    Exited (0) 10 minutes ago    lucid_darwin
b2a608dc3     nginx     "/docker-entrypoint..." 25 minutes ago    Exited (0) 16 minutes ago    keen_leakey
ubuntu@ip-172-31-45-84:~$

```

- Inspect the NGINX Image
- View detailed information about the image:

```
sudo docker image inspect nginx
```

```
ubuntu@ip-172-31-45-84:~$ sudo docker image inspect nginx
[
  {
    "Id": "sha256:ec4ed8b5299e5e90694af7750eb6dffd2627317d30544d056b0371f8082f7bce",
    "RepoTags": [
      "nginx:latest"
    ],
    "RepoDigests": [
      "nginx@sha256:ec4ed8b5299e5e90694af7750eb6dffd2627317d30544d056b0371f8082f7bce"
    ],
    "Comment": "buildkit.dockerfile.v0",
    "Created": "2026-06-24T01:21:58.377973076Z",
    "Config": {
      "ExposedPorts": {
        "80/tcp": {}
      },
      "Env": [
        "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
        "NGINX_VERSION=1.31.2",
        "NJS_VERSION=0.9.9",
        "NJS_RELEASE=1-trixie",
        "ACME_VERSION=0.4.1",
        "PKG_RELEASE=1-trixie",
        "DYNPKG_RELEASE=1-trixie"
      ],
      "Entrypoint": [
        "/docker-entrypoint.sh"
      ],
      "Cmd": [
        "nginx",
        "-g",
        "daemon off;"
      ],
      "Labels": {
        "maintainer": "NGINX Docker Maintainers <docker-maint@nginx.com>"
      }
    }
  ]
}
```

- View Image History (Layers)
- Display the layers that make up the image:
- `sudo docker history nginx`

```
ubuntu@ip-172-31-45-84:~$ sudo docker history nginx
IMAGE          CREATED          CREATED BY                                      SIZE      COMMENT
ec4ed8b5299e   13 days ago     CMD ["nginx" "-g" "daemon off;"]              0B        buildkit.dockerfile.v0
<missing>      13 days ago     STOPSIGNAL SIGQUIT                            0B        buildkit.dockerfile.v0
<missing>      13 days ago     EXPOSE map[80/tcp:{}]                         0B        buildkit.dockerfile.v0
<missing>      13 days ago     ENTRYPOINT ["/docker-entrypoint.sh"]          0B        buildkit.dockerfile.v0
<missing>      13 days ago     COPY 30-tune-worker-processes.sh /docker-ent... 16.4kB    buildkit.dockerfile.v0
<missing>      13 days ago     COPY 20-envsubst-on-templates.sh /docker-ent... 12.3kB    buildkit.dockerfile.v0
<missing>      13 days ago     COPY 15-local-resolvers.envsh /docker-entryp... 12.3kB    buildkit.dockerfile.v0
<missing>      13 days ago     COPY 10-listen-on-ipv6-by-default.sh /docker... 12.3kB    buildkit.dockerfile.v0
<missing>      13 days ago     COPY docker-entrypoint.sh / # buildkit        8.19kB    buildkit.dockerfile.v0
<missing>      13 days ago     RUN /bin/sh -c set -x && groupadd --syst...    87.1MB    buildkit.dockerfile.v0
<missing>      13 days ago     ENV DYNPKG_RELEASE=1-trixie                    0B        buildkit.dockerfile.v0
<missing>      13 days ago     ENV PKG_RELEASE=1-trixie                       0B        buildkit.dockerfile.v0
<missing>      13 days ago     ENV ACME_VERSION=0.4.1                         0B        buildkit.dockerfile.v0
<missing>      13 days ago     ENV NJS_RELEASE=1-trixie                       0B        buildkit.dockerfile.v0
<missing>      13 days ago     ENV NJS_VERSION=0.9.9                         0B        buildkit.dockerfile.v0
<missing>      13 days ago     ENV NGINX_VERSION=1.31.2                       0B        buildkit.dockerfile.v0
<missing>      13 days ago     LABEL maintainer=NGINX Docker Maintainers <d... 0B        buildkit.dockerfile.v0
<missing>      2 weeks ago     # debian.sh --arch 'amd64' out/ 'trixie' '@1... 87.4MB    debuerreotype 0.17
ubuntu@ip-172-31-45-84:~$
```

- Stop the container and remove the containers and images
- Run the command

```
sudo docker ps -aq
```

```
sudo docker rm -f $(sudo docker ps -aq)
```

```
ubuntu@ip-172-31-45-84:~$ sudo docker ps -aq
280ea934ca39
2198d760ba3f
1b30746055aa
b2a608dcede3
ubuntu@ip-172-31-45-84:~$ sudo docker rm -f $(sudo docker ps -aq)
280ea934ca39
2198d760ba3f
1b30746055aa
b2a608dcede3
ubuntu@ip-172-31-45-84:~$ sudo docker ps -aq
ubuntu@ip-172-31-45-84:~$
```

- To remove the image, run the commands

```
sudo docker images
```

```
sudo docker rmi nginx
```

```
ubuntu@ip-172-31-45-84:~$ sudo docker images
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
nginx:latest ec4ed8b5299e 241MB 66MB
ubuntu@ip-172-31-45-84:~$ sudo docker rmi nginx
Untagged: nginx:latest
Deleted: sha256:ec4ed8b5299e5e90694af7750eb6dfffd2627317d30544d056b0371f8082f7bce
ubuntu@ip-172-31-45-84:~$ sudo docker images
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
ubuntu@ip-172-31-45-84:~$
```

Info → U In Use

Info → U In Use